

DevOps Fundamentals Mini Project

Project Title: Cloud-Based Linux Server Deployment with
GitHub Version Control and Automation.

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Project Overview

This project was completed as part of the ITVedant DevOps Fundamentals course. The goal of the project was to understand and implement the fundamental concepts of DevOps by deploying a website on an AWS Ubuntu server.

The project covers AWS Cloud infrastructure, Linux server administration, Git & GitHub version control, Bash scripting, Apache web server configuration, and Linux log monitoring.

The implementation simulates the daily responsibilities of a Junior DevOps Engineer working in an organization.

Project Objectives

- Configure AWS Cloud infrastructure.
- Create a custom VPC and Public Subnet.
- Configure Internet Gateway and Route Table.
- Launch an Ubuntu EC2 Instance.
- Install required Linux packages.
- Deploy a website using Apache Web Server.
- Use Git and GitHub for version control.
- Develop Bash automation scripts.
- Monitor Linux system logs.
- Prepare professional technical documentation.

AWS Infrastructure Setup

AWS provides the cloud environment where the website is hosted.

Step 1 – Create Custom VPC

A Virtual Private Cloud (VPC) provides an isolated virtual network for AWS resources.

Step 2 – Create Public Subnet

A Public Subnet was created inside the VPC.

Step 3 – Configure Internet Gateway

An Internet Gateway was attached to the VPC.

Step 4 – Configure Route Table

Configure Route Table to Enables internet access from the public subnet.

Step 5 – Configure Security Group

Security Groups act as virtual firewalls.

- SSH for server login.
- HTTP for website access.

Step 6 – Launch Ubuntu EC2 Instance

Provides a Linux server to host the application.

Step 7 – Connect Using SSH

Connected to the Ubuntu server using SSH provides secure remote access to the Linux server.

Ubuntu Server Configuration

After connecting to the EC2 instance, the server was configured.

System Update

```
#sudo apt update -y
```

System Upgrade

```
#sudo apt upgrade -y
```

Install Apache2

```
#sudo apt install apache2 -y
```

Install Curl

```
#sudo apt install curl -y
```

Install Vim

```
#Sudo apt install vim -y
```

Verify Installation

```
#git --version
```

```
#apache2 -v
```

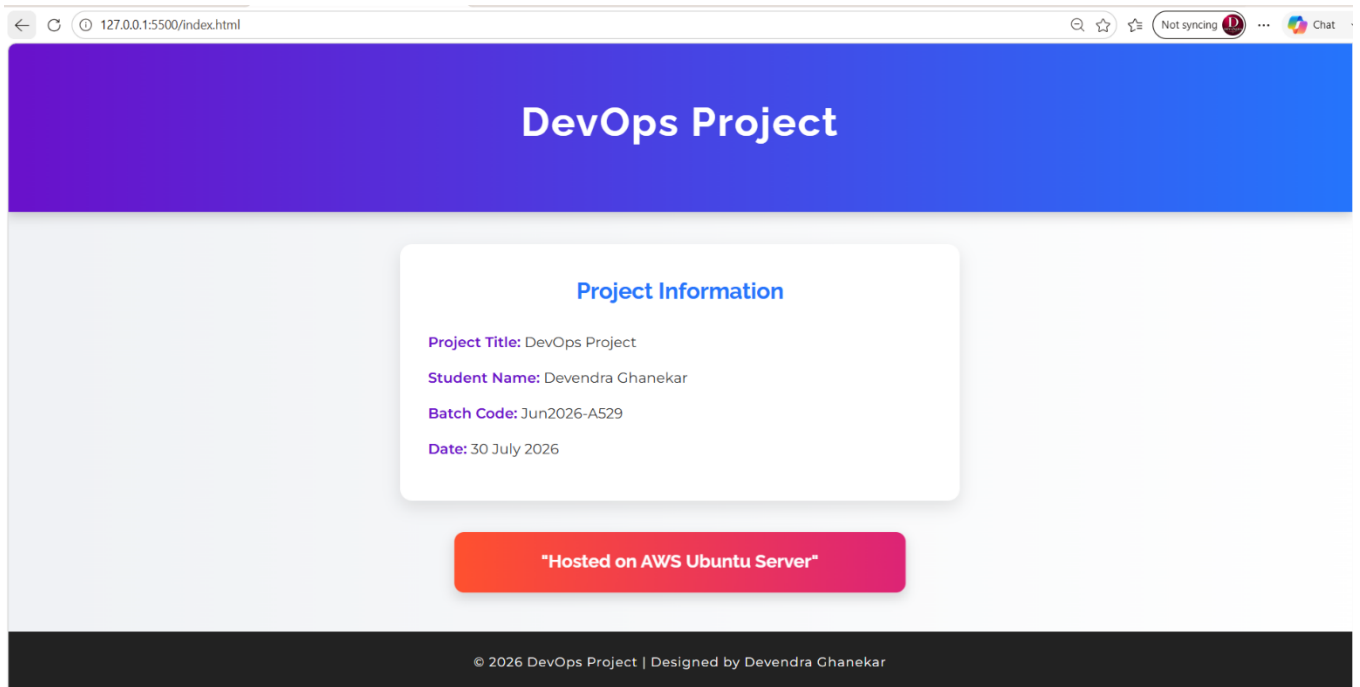
```
#curl --version
```

```
#vim --version
```

```
#sudo systemctl status apache2
```

Website Deployment

A simple HTML webpage was created containing.



Git & GitHub Implementation

A GitHub repository name **DevOps-Fundamentals-Project** was created.

Git commands are used

```
#git init
```

```
#git remote -v
```

```
#git remote add origin
```

```
#git add .
```

```
#git commit -m "new project deploy"
```

```
#git branch
```

```
#git push -u origin main
```

Bash Automation Scripts

1.Health-check.sh :

- CPU Usage
- Memory Usage
- Disk Usage
- Logged-in Users
- Running Processes
- Server Uptime

2. Apache-monitor.sh :

- Checks Apache service.
- Restarts Apache automatically if stopped.
- Displays a confirmation message.

3.Backup.sh :

- Compress website files.
- Create timestamp.
- Store backup inside Backup folder.

Conclusion

This project provided practical experience in cloud infrastructure, Linux server administration, version control, automation, and website deployment. I successfully created a secure AWS environment, configured an Ubuntu server, deployed a website using Apache, managed the project with Git and GitHub, developed Bash automation scripts, and monitored Linux logs.

The project helped me understand how different DevOps tools and technologies work together in a real-world environment and strengthened my skills for a Junior DevOps Engineer role.